

EmbeddingBag#

<https://pytorch.org/docs/stable/generated/torch.nn.quantized.EmbeddingBag>

Description:

A quantized EmbeddingBag module with quantized packed weights as inputs. We adopt the same interface as torch.nn.EmbeddingBag, please see

<https://pytorch.org/docs/stable/generated/torch.nn.EmbeddingBag.html> for documentation. Similar to EmbeddingBag, attributes will be randomly initialized at module creation time and will be overwritten later weight (Tensor) – the non-learnable quantized weights of the module of shape (num_embeddings, embedding_dim) (\text{num_embeddings}, \text{embedding_dim})(num_embeddings, embedding_dim). Create a quantized embedding_bag module from a float module mod (Module) – a float module, either produced by torch.ao.quantization utilities or provided by user

Function Signature

```
class torch.nn.quantized.EmbeddingBag(num_embeddings,
embedding_dim, max_norm=None, norm_type=2.0,
scale_grad_by_freq=False, mode='sum', sparse=False,
_weight=None, include_last_offset=False, dtype=torch.qint8)
[source]#
```

Variables

Examples::

```
classmethod from_float(mod,
use_precomputed_fake_quant=False)[source]#
```

Parameters

Code Examples

```
>>> m = nn.quantized.EmbeddingBag(num_embeddings=10,
embedding_dim=12, include_last_offset=True, mode='sum')
>>> indices = torch.tensor([9, 6, 5, 7, 8, 8, 9, 2, 8, 6, 6, 9,
1, 6, 8, 8, 3, 2, 3, 6, 3, 6, 5, 7, 0, 8, 4, 6, 5, 8, 2, 3])
>>> offsets = torch.tensor([0, 19, 20, 28, 28, 32])
>>> output = m(indices, offsets)
>>> print(output.size())
torch.Size([5, 12])
```

Detailed Documentation

Documentation

A quantized EmbeddingBag module with quantized packed weights as inputs. We adopt the same interface as `torch.nn.EmbeddingBag`, please see <https://pytorch.org/docs/stable/generated/torch.nn.EmbeddingBag.html> for documentation.

Similar to `EmbeddingBag`, attributes will be randomly initialized at module creation time and will be overwritten later

`weight` (Tensor) – the non-learnable quantized weights of the module of shape $(\text{num_embeddings}, \text{embedding_dim})$ (`num_embeddings`, `embedding_dim`)

Create a quantized `embedding_bag` module from a float module

`mod` (Module) – a float module, either produced by `torch.ao.quantization` utilities or provided by user